

CYBER SECURITY AND ETHICAL HACKING

Question Bank

Part-A

1. Define the need of cyber security.
2. Mention any two objectives of cyber security.
3. Differentiate cyber security and network security.
4. Define salami attack with example.
5. Write a short note on email bombing.
6. Describe the concept of botnet.
7. Differentiate between virus and Trojan (any two points).
8. Briefly write what happens at the stages of reconnaissance.
9. Write the name of any two tools used in passive and active attack.
10. Mention any two tools used in mobile forensics.
11. Define metadata in the context of digital evidence.
12. Provide one example of digital signature–related crime.
13. What is the main motive behind the Email Spoofing crime?
14. Provide the types of digital crime related to section 66A and 66B.
15. Write down two points on the necessity of using IT act 2000.
16. Write two importance of cyber security.
17. Give an example of salami attack.
18. Give an example of online social engineering attack.
19. Write any two uses of steganography in cyber crime.
20. Differentiate between symmetric and asymmetric cryptography (any two points).
21. Name any two techniques to prevent spyware attacks.
22. Write down the concept of exploitation in ethical hacking.
23. Specify the role of Trojan in cyber crime.
24. Specify the name of two tools used in cyber forensic.
25. Describe the concept of dumpster diving with example.
26. Give two example of online financial fraud.
27. Give one reason why digital signatures are used in online transactions.
28. Give an example of attack vector and botnet.
29. Provide the types of digital crime related to section 67 and 67B.
30. Define cyber stalking and mention one example.
31. What is the relationship between cybercrime and information security?
32. What is an attack vector?
33. Mention two security challenges in mobile devices.
34. Define credit card fraud in mobile computing.
35. What is authentication service security?

36. What is a proxy server?
37. Define keylogger and spyware.
38. What is the purpose of the scanning phase in ethical hacking?
39. Name any two tools used during the exploitation phase.
40. Define chain of custody in digital forensics.

Part-B

1. Give two example of online social engineering attack.
2. How does SQL injection attacks occur, explain with suitable example.
3. Explain packet sniffing and how attackers use it.
4. Discuss the role of the Metasploit Framework in ethical hacking.
5. Explain the importance of chain of custody with an example.
6. Describe any two forensic investigation tools and their uses.
7. Describe MITM attack and stages of ransomware with example.
8. Write down an example of buffer overflow attack and its remedial steps.
9. Give two examples where phishing attack is done through search engine.
10. Explain how criminals attack banking systems using different cyber techniques.
11. Describe buffer overflow attacks with an example.
12. Explain post-exploitation with examples.
13. Describe some challenges in network forensics.
14. Analyze how attackers use multiple tools—phishing and SQL injection and spyware—in coordinated cyber attacks.
15. Evaluate how enterprise mobility policies can mitigate these risks of proliferation of mobile devices.
16. Transmitting malicious text or images to another person without the owner's permission falls under which section category of cyber law.
17. Explain the various types of cyber offenses related to mobile and wireless devices.
18. Discuss the steps of ethical hacking (Planning, Reconnaissance, Scanning, Exploitation, Post-Exploitation, Reporting).
19. Describe different types of DoS and DDoS attacks with examples.
20. Explain the Digital Forensics Life Cycle with all phases and tools used.

Part-C

1. *In an RSA cryptosystem, Alice selects two prime numbers $p = 19$ and $q = 71$ to generate her public and private keys. If Alice's public key exponent is $e = 11$, compute: Her private key (d), The ciphertext when she sends plaintext = 120.*

2. *Explain the schematic diagram of steganography. Convert the plain text (CYBER SECURITY) to cipher text by using mono alphabetic cipher algorithm.*
3. *Describe how botnets are used to perform DDoS attacks. Explain with an example how multiple compromised systems target a single server.*
4. *Explain all phases of ethical hacking (Planning, Reconnaissance, Scanning, Exploitation, Post-Exploitation, Reporting) with diagrams and tools.*
5. *A bank customer complains that ₹50,000 was transferred from his account without authorization. Describe how cyber forensics can be used to trace the transaction, identify logs, and detect the criminal.*
6. *Explain the RSA Algorithm in detail, including key generation, encryption, decryption, mathematical principles, and an example using public key ($e = 17$, $n = 33$) and private key ($d = 3$, $n = 33$).*
7. *Explain the Caesar Cipher Algorithm. If Alice sends the message "TECHNO INDIA UNIVERSITY" using a secret key 19, what ciphertext will Bob receive. After decryption, what plaintext will Bob get.*
8. *Explain different categories of SQL Injection payloads and show how attackers extract database records.*
9. *Explain the complete Digital Forensics Life Cycle with all its phases (Preparation/Identification, Collection, Examination, Analysis, Reporting, and Presentation). Discuss the key activities, tools, and challenges in each phase, with a real-world example of investigating a corporate data breach.*
10. *Describe various security challenges posed by mobile and wireless devices and explain organizational measures to mitigate them.*
11. *Explain cyber stalking, botnets, attack vectors, and cloud-based cyber offenses with suitable real-life examples.*
12. *Discuss in detail phishing, password cracking, keyloggers, spyware, and identity theft techniques with examples.*
13. *Describe the chain of custody, digital evidence handling, email forensics, forensics auditing, and anti-forensics techniques in detail.*
14. *Explain cyber stalking, botnets, attack vectors, and cloud-based cyber offenses with suitable real-life examples.*